

DeXposure-Claw: An Agentic System for Risk Monitoring and Decision Recommendation on Decentralised Financial Networks

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Abstract

Decentralized finance exposes supervisors to fast-moving, networked credit risks in which shocks can propagate across protocols before human operators can inspect raw on-chain data. We introduce DeXposure-Claw, a forecast-grounded agentic supervision system that routes LLM decisions through structured evidence rather than raw transactions. This addresses a problem in the existing LLM-agent approaches that are poorly suited to this setting because they can over-read weak or stale evidence and recommend high-stakes interventions without a regulator-aligned way to measure false alarms. The system uses DeXposure-FM, a graph time-series foundation model, to forecast future exposure networks, converts those forecasts into monitor alerts and stress-scenario evidence, and applies data-health and confidence gates before emitting auditable supervisory tickets with rationales. On five years of weekly DeFi exposure graphs, DeXposure-Claw improves ticket F1 over a conservative rules baseline from 0.0076 to 0.0233, while a cheaper Sonnet 4.6 decision model further raises F1 to 0.0288, eliminates measured false interventions, and directionally improves explanation quality, positioning the system as a recall-explanation Pareto option for human-in-the-loop DeFi supervision rather than a replacement for conservative rules.

1 Introduction

Decentralized finance (DeFi) creates fast-moving networks of token-mediated credit exposure among lending protocols, decentralized exchanges, stablecoins, bridges, and yield aggregators. Recent crises such as Terra/Luna, FTX, and SVB/USDC show that shocks can propagate across this network before supervisors can manually inspect raw on-chain data (Vidal-Tomás et al., 2023). This motivates decision-support systems that can forecast emerging exposure risk, identify affected protocols, and recommend timely supervisory responses.

A straightforward approach is to build an LLM agent that reasons directly over on-chain measurements. We argue that this is unsafe for high-stakes supervision: general-purpose LLM agents may produce plausible rationales while over-reading incomplete, stale, or weak evidence, thereby triggering unnecessary high-severity interventions. Existing systemic-risk evaluations also often rank protocols by fractional exposure changes (Bertomeu et al., 2024; Gonon et al., 2025; Li et al., 2025), which can overemphasize small protocols and fail to reflect regulator-relevant loss priorities.

We introduce *DeXposure-Claw*, a forecast-grounded agentic system for DeFi risk monitoring and decision recommendation. Rather than asking an LLM to reason over raw transactions, the system routes decisions through structured forecasted evidence. A graph time-series foundation model forecasts future exposure networks; deterministic monitors and stress scenarios convert these forecasts into alerts, attribution information, scenario losses, uncertainty estimates, and data-health scores; an LLM then emits ranked supervisory tickets with targets, severities, and evidence-linked rationales. Data-health and confidence gates restrict intervention-level recommendations, and every ticket is released with its evidence bundle, rationale, and gate states for auditability.

We evaluate the system with DeXposure-Bench, a six-axis harness covering forecast quality, warning behavior, uncertainty calibration, stress-scenario fidelity, ticket quality, and robustness. Its decision axis uses a regulator-aligned absolute-loss ground truth, allowing false interventions and supervisory relevance to be measured directly. Across five years of weekly DeFi exposure graphs and eight reference implementations, the full DeXposure-Claw stack improves ticket F1 from 0.0076 for a conservative persistence-rules baseline to 0.0233. Replacing the main Opus 4.7 decision model with Sonnet 4.6 yields the strongest

operating point, raising F1 to 0.0288, reducing measured false interventions to zero, and directionally improving explanation quality at lower inference cost. These results position DeXposure-Claw as an auditable recall–explanation Pareto option for human-in-the-loop DeFi supervision, rather than a replacement for conservative rule-based systems.

2 Related Work

Bench positioning and ground-truth definition. LLM-agent benchmarks (HELM (Liang et al., 2022), SWE-bench (Jimenez et al., 2024), Agent-Bench (Liu et al., 2024)) score open-ended reasoning, software repair, and generic agent behaviour; temporal-graph benchmarks (TGB (Huang et al., 2023), OGB (Hu et al., 2020)) score structural prediction quality. Neither scores whether an LLM agent’s supervisory decisions match what a regulator would actually prioritise. The closest prior systemic-risk evaluations (Bertomeu et al., 2024; Gonon et al., 2025; Li et al., 2025) rank protocols by *fractional* weight change, which disproportionately surfaces tiny protocols of low systemic relevance. DeXposure-Bench replaces this with an *absolute-loss* ground truth (§3).

LLM agents in finance and DeFi. General agent patterns such as ReAct (Yao et al., 2023) combine reasoning with tool use; FinGPT (Yang et al., 2023) adapts language models to financial data. Within DeFi, agentic systems have so far targeted transaction auditing (Yao et al., 2026), intent mining (Mao et al., 2025), smart-contract verification (Hu et al., 2026; Kong et al., 2026), price manipulation (Liu et al., 2025), and explanation for graph anomalies (Watson et al., 2025); portfolio-oriented agents combine graph encoders with LLMs (Luo et al., 2025; Jeon and Lee, 2026). These systems consume raw transactions or token text, not structured forecasted evidence, and none reports a false-intervention rate against a regulator-aligned ground truth—a gap our characterisation (§6) closes.

Forecast-grounded LLM agents in other domains. Pairing a domain forecaster with an LLM decision layer is an emerging deployment pattern. In macroeconomics, ChatGPT-augmented PMI nowcasting (de Bondt and Sun, 2025) and LLM-driven macroeconomic forecasting (Carriero et al., 2025) both feed structured numeric inputs into an LLM that produces interpretable narratives; a BIS primer surveys the wider pattern (Kwon

et al., 2024). Time-series foundation models such as Chronos (Ansari et al., 2024), Lag-Llama (Rasul et al., 2024), and TimesFM (Das et al., 2024) make the forecaster reusable across tasks; tabular foundation models extend the same idea to heterogeneous structured data (Hollmann et al., 2025; Ereemeev et al., 2025). None of these forecast→LLM pipelines, to our knowledge, has been scored against a regulator-aligned ground truth for high-stakes financial-network supervision.

3 Preliminaries

We model decentralized-finance credit exposure as a temporal weighted directed graph. Let $\mathcal{P}$ be the universe of protocols and $\mathcal{X}$ the universe of tokens. At time t , protocol p holds tokens $X_p(t) \subseteq \mathcal{X}$, while protocol q issues or is economically linked to tokens $X_G(q) \subseteq \mathcal{X}$. Protocol p is exposed to q if

$$X_p(t) \cap X_G(q) \neq \emptyset. \quad (1)$$

The exposure network is

$$G_t = (V_t, E_t, W_t), \quad (2)$$

where $V_t \subseteq \mathcal{P}$ is the active protocol set, $E_t \subseteq V_t \times V_t$ is the directed exposure set, and $W_t[p, q] = w_{pq,t} \geq 0$ is the USD exposure from p to q . For protocol v , define its incident exposure mass as

$$w_t(v) = \sum_{u \in V_t} w_{uv,t} + \sum_{u \in V_t} w_{vu,t}. \quad (3)$$

The observed history is a sequence of snapshots $\mathcal{G}_{1:T} = \{G_1, \dots, G_T\}$, optionally with covariates X_t .

At decision epoch t , the forecasting problem is to estimate G_{t+h} for horizons $h \in \mathcal{H}$. A probabilistic temporal-graph forecaster outputs

$$P_{t,h} = P(G_{t+h} \mid G_{\leq t}, X_{\leq t}). \quad (4)$$

For each ordered pair (p, q) , let

$$\pi_{pq,t+h} = \Pr(e_{pq,t+h} = 1 \mid G_{\leq t}, X_{\leq t}), \quad (5)$$

and

$$\mu_{pq,t+h} = \mathbb{E}[w_{pq,t+h} \mid e_{pq,t+h} = 1, G_{\leq t}, X_{\leq t}]. \quad (6)$$

A representative predicted graph $\hat{G}_{t+h}$ is constructed by

$$\begin{aligned} \hat{E}_{t+h} &= \{(p, q) : \pi_{pq,t+h} > \pi_{\min}\}, \\ \hat{W}_{pq,t+h} &= \pi_{pq,t+h} \mu_{pq,t+h}. \end{aligned}$$

Uncertainty is represented by Monte Carlo samples

$$\tilde{G}_{t+h}^{(1)}, \dots, \tilde{G}_{t+h}^{(M)} \sim P_{t,h}. \quad (7)$$

A monitor maps graphs to scalar risk statistics $\Phi_j : \mathcal{G} \rightarrow \mathbb{R}$, such as maximum PageRank, HHI concentration, density, PageRank Gini, or degree Gini. Given a rolling window of length L , define

$$\mu_{j,t} = \frac{1}{L} \sum_{\ell=1}^L \Phi_j(G_{t-\ell}),$$

$$\sigma_{j,t} = \sqrt{\frac{1}{L-1} \sum_{\ell=1}^L (\Phi_j(G_{t-\ell}) - \mu_{j,t})^2}.$$

An alert fires when

$$A_{j,t,h} = \mathbf{1} \left\{ \frac{|\Phi_j(\tilde{G}_{t+h}) - \mu_{j,t}|}{\max(\sigma_{j,t}, \epsilon)} > z \right\}. \quad (8)$$

A stress scenario is a graph perturbation $s : \mathcal{G} \rightarrow \mathcal{G}$. Let G^s be the shocked graph and define

$$\text{Mass}(G) = \sum_{e \in E(G)} w_e,$$

$$L(s; G) = \frac{\text{Mass}(G) - \text{Mass}(G^s)}{\text{Mass}(G)}.$$

For sampled graphs, let $\ell_m(s, h) = L(s; \tilde{G}_{t+h}^{(m)})$. If $\ell_{(1)} \geq \dots \geq \ell_{(M)}$ are sorted losses, then

$$\text{CVaR}_\lambda(s, h) = \frac{1}{\lceil \lambda M \rceil} \sum_{m=1}^{\lceil \lambda M \rceil} \ell_{(m)}. \quad (9)$$

For evaluation, we define regulator-aligned ground truth by absolute exposure loss. For protocol v and horizon h ,

$$\Delta_t^h(v) = w_t(v) - w_{t+h}(v). \quad (10)$$

Given percentile $\pi \in (0, 1)$, the stressed set is

$$S_t^h = \text{top}_\pi \left\{ v : w_t(v) > 0, \Delta_t^h(v) > 0 \right\}, \quad (11)$$

where protocols are ranked by $\Delta_t^h(v)$.

A supervisory system emits tickets

$$D_t = \{\tau_{t,1}, \dots, \tau_{t,K_t}\}, \quad \tau_{t,k} = (u_{t,k}, Y_{t,k}, r_{t,k}), \quad (12)$$

where $u_{t,k}$ is an action, $Y_{t,k} \subseteq V_t$ is a target set, and $r_{t,k}$ is a rationale. The action space is

$$\mathcal{U} = \{\text{MONITOR, INVESTIGATE, RECOMMENDREDUCE, CONTINGENCY}\}.$$

For predicted targets $\hat{S}_t$, ticket quality is

$$\text{Precision} = \frac{|\hat{S}_t \cap S_t^h|}{|\hat{S}_t|},$$

$$\text{Recall} = \frac{|\hat{S}_t \cap S_t^h|}{|S_t^h|},$$

$$\text{F1} = \frac{2 \text{ Precision Recall}}{\text{Precision} + \text{Recall}}.$$

Let $\hat{S}_t^{\text{int}}$ be targets of intervention-level actions in $\{\text{RECOMMENDREDUCE, CONTINGENCY}\}$. The false-intervention rate is

$$\text{FIR} = \frac{|\hat{S}_t^{\text{int}} \setminus S_t^h|}{|\hat{S}_t^{\text{int}}|}. \quad (13)$$

Finally, define the data-health score

$$\text{DH}_t = \frac{1}{4} (\text{fresh}_t + \text{miss}_t + \text{topo}_t + \text{disc}_t). \quad (14)$$

For alert a at horizon h , with Monte Carlo coefficient of variation $\text{CV}_{t,h}(a)$, define

$$C_t(a) = \text{DH}_t \cdot \frac{1}{1 + \text{CV}_{t,h}(a)} \cdot \frac{1}{1 + \log(1 + h)}. \quad (15)$$

If $\bar{C}_t$ is the mean active-alert confidence, intervention-level actions are feasible only when

$$\text{DH}_t \geq \tau_{\text{data}} \quad \text{and} \quad \bar{C}_t \geq \tau_{\text{conf}}. \quad (16)$$

4 Pipeline

DeXposure-Claw is a four-layer pipeline for forecast-grounded DeFi supervision (Figure 1). Layer 1 uses DeXposure-FM to forecast near-future credit-exposure graphs. Layer 2 converts the forecast distribution into a typed evidence bundle containing monitor alerts, uncertainty estimates, and stress-scenario losses. Layer 3 passes this bundle to an LLM, which emits ranked supervisory tickets with targets, severities, and rationales. Layer 4 applies data-health and confidence gates to determine whether intervention-level actions are admissible. The pipeline runs at weekly cadence on a single RTX 4090, with well under a dollar of LLM API spend per weekly decision (a full DeXposure-Bench leaderboard run is approximately \$10).

4.1 Forecaster interface

At each decision epoch t , the agent observes a credit-exposure snapshot $G_t = (V_t, E_t, W_t)$, an

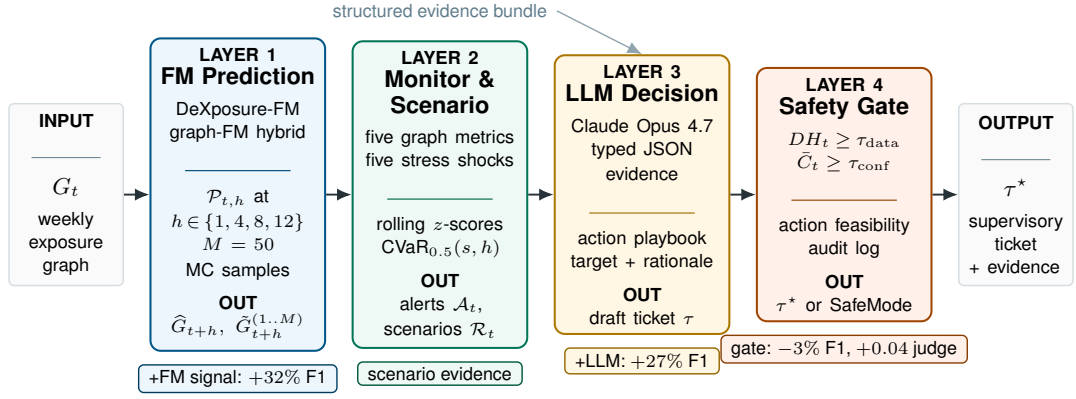


Figure 1: DeXposure-Claw: a four-layer supervisory pipeline. Given a weekly exposure graph G_t , the system first forecasts future exposure networks with a foundation model (Layer 1), converts those forecasts into monitor alerts and stress-scenario evidence (Layer 2), asks an LLM to produce supervisory tickets from the typed evidence bundle (Layer 3), and applies safety gates before releasing either a ticket τ^* or a safe-mode notice (Layer 4). Bottom annotations report the per-layer F1 and judge-score lifts from Section 6.

inter-protocol token-mediated exposure graph defined in Appendix A, together with node covariates X_t . For each forecast horizon $h \in \mathcal{H} = \{1, 4, 8, 12\}$ weeks, it queries DeXposure-FM for a predictive distribution

$$\mathcal{P}_{t,h} \leftarrow \text{ForecastFM}(G_t, X_t, h). \quad (17)$$

From this distribution, the agent constructs an expected-weight predicted graph $\hat{G}_{t+h}$ by thresholding edge-existence probabilities at $\pi_{\min} = 0.2$. It also draws $M = 50$ Monte Carlo samples to quantify predictive uncertainty. Full predicted-graph construction equations are given in Appendix A.

4.2 Structured evidence

Layer 2 converts $\hat{G}_{t+h}$ and its Monte Carlo samples into a typed evidence bundle. Five *monitor functionals* summarize concentration, fragility, and instability in the predicted graph: top-1 PageRank (Brin and Page, 1998), HHI concentration (Rhoades, 1993), network density, PageRank Gini, and degree Gini. An alert fires when a monitor statistic deviates from its $L = 42$ week rolling baseline by more than $z = 2$ standard deviations. Each alert includes two additional fields: (i) an attribution top- K list of contributing edges, ranked by marginal contribution, and (ii) an uncertainty-aware confidence score $C_t(a) \in [0, 1]$ that decreases with lower data health, wider Monte Carlo dispersion, and longer forecast horizons.

In parallel, a *scenario engine* applies five standardized stress shocks to both $\hat{G}_{t+h}$ and each Monte Carlo sample: single protocol failure, bridge

cluster failure, stablecoin de-peg, sector lending shock, and correlated top-10 stress. For each scenario s and horizon h , the engine aggregates the worst-half sampled losses as $\text{CVaR}_{0.5}(s, h)$. Full monitor and scenario definitions are provided in Appendix B.

4.3 LLM decision layer

Layer 3 receives the evidence bundle as a typed JSON payload containing alerts with attribution top- K , a scenario-CVaR table, Monte Carlo dispersion summaries, and the scalar data-health score DH_t . The LLM is prompted to emit a ranked list of supervisory tickets. Each ticket contains (a) a severity level from a fixed four-action playbook (Monitor, Investigate, Recommend-Reduce, and Contingency), (b) a target set of protocols, and (c) a free-form rationale that cites specific evidence fields by name. To measure self-consistency, we repeat the decision call three times at temperature 0 and report the Jaccard overlap of ticket targets and severities.

4.4 Safety gates

Layer 4 determines whether intervention-level recommendations are feasible. The *data-health gate* computes

$$\text{DH}_t = \text{mean}(\text{fresh}_t, \text{miss}_t, \text{topo}_t, \text{disc}_t). \quad (18)$$

If $\text{DH}_t < \tau_{\text{data}} = 0.7$, the system enters safe mode and emits only Monitor or Investigate tickets, regardless of the LLM’s recommended severity. The *confidence gate* blocks intervention-level tickets whenever the mean alert confidence satisfies

$\bar{C}_t < \tau_{\text{conf}} = 0.6$, even if the data-health gate passes. Every emitted ticket carries the full evidence bundle, the LLM rationale, and the gate states that produced it. The resulting audit log is the unit of release (Section 7).

The end-to-end procedure is given in Appendix B (Algorithm 1).

5 Evaluation Harness

Why a new harness. Scoring an LLM-agent supervisory pipeline requires evidence on six independent axes that no existing benchmark covers together: how good is the forecast (b1_forecast), how early does the monitor turn (b2_warning), is the predictive uncertainty calibrated (b3_calibration), do scenario losses match reality (b4_stress), do the emitted tickets match what a regulator would prioritise (b5_decision), and what happens under degraded data (b6_robustness). No existing benchmark covers these axes together (§2); DeXposure-Bench unifies them on a shared dataset and reference-method set so layer-wise ablations can be reported independently.

Dataset and split. We use the DeXposure dataset of weekly directed weighted exposure graphs (4,300 protocols, 24,300 tokens, 43.7M exposure entries, 283 weekly snapshots from 2020-03 to 2025-08 (Wu et al., 2025)). We adopt a strict no-leakage temporal split: 2020-03–2024-06 train (~ 222 weeks), 2024-07–2024-12 validation (~ 26 weeks, used for conformal calibration and threshold tuning), 2025-01–2025-08 test (~ 33 weeks, frozen leaderboard). All DeXposure-FM fine-tuning and model selection likewise use pre-2025 data only (§A). For b2_warning we additionally run a historical event study around Terra/Luna (2022-05-09), FTX (2022-11-07), and SVB/USDC (2023-03-10); the event study is reported separately from the 2025 predictor-specific leaderboard.

Regulator-aligned absolute-loss ground truth. Prior systemic-risk evaluations rank protocols by *fractional* weight change, which silently up-weights tiny protocols of low systemic relevance. DeXposure-Bench ranks instead by *absolute* loss, scoring tickets against the stressed set $\mathcal{S}_t^h$ and FIR of §3. We fix $\pi = 0.05$ and the ground-truth horizon $h = 4$ weeks throughout (yielding $|\mathcal{S}_t^h| \approx 283$ protocols per week); the last four test

weeks lack a 4-week-ahead snapshot, so decision axes score $n = 29$ weeks. The same ground truth anchors b2_warning, b5_decision, and the LLM-decision evaluation pipeline.

Eight reference implementations. The harness bundles eight reference methods spanning the four ablation cuts of the four-layer framework: heuristic streaming monitor (h1), persistence+rules (m1), snapshot LLM (m2), EvolveGCN (Pareja et al., 2020) (m3), DeXposure-FM-only (m4), DeXposure-FM+rules (m5), DeXposure-FM+LLM (m6), and DeXposure-FM+LLM+gate (m7). Each method exposes the same predict_graph or decide_actions interface so adding a ninth method is a single-file change. The full method specification and per-axis schema are in §C–§D.

Reproducibility. A complete leaderboard run finishes in under four GPU-hours on a single RTX 4090 plus approximately \$10 of LLM API spend, making the harness accessible without industrial compute. The harness fixes seeds, snapshots library versions, and hashes checkpoints; every reported number reproduces bit-exactly from the released artefacts.

6 Empirical Results

We use DeXposure-Bench to ask three questions. Does DeXposure-FM provide forecast signal that persistence does not (b1, b3, b6)? Does routing the LLM through structured evidence improve supervisory tickets once precision, recall, explanation quality, and false interventions are scored jointly (b4, b5)? Which components are load-bearing, and which are reserves under degradation (ablations)?

DeXposure-FM contributes trend, calibration, and robustness; persistence still wins on static point error. On static error metrics, persistence is a deceptively strong baseline: at $h = 4$ it beats the FM on PageRank MAE (3.4×10^{-5} vs 4.5×10^{-5}) and edges it on Spearman rank correlation (0.570 vs 0.558). The FM contributes signal where persistence is structurally pinned: trend consistency 0.525–0.632 across horizons (persistence: 0); b3_calibration PI coverage 0.913 against a 0.90 target and ECE 0.013; b6_robustness mean relative degradation 0.113 vs 0.148 for persistence (24% less). The full per-horizon and per-regime audits are in §E.

Headline decision-quality result. Table 1 consolidates the central b5_decision comparison. The persistence+rules baseline (m1) achieves the highest single-axis precision (0.720); like m2/m5 it never escalates beyond Investigate, so its FIR is undefined, not a safety guarantee. Against this baseline, the full DeXposure-FM+LLM+gate stack (m7) lifts ticket F1 from 0.0076 to 0.0233 (+207%; $p < 10^{-4}$). Against the comparable pure-LLM baseline that consumes raw snapshots without the forecaster (m2, the only other LLM-bearing row), m7 lifts LLM-as-judge explanation quality from 2.24 to 2.45 on a 1–5 scale (+9.4%) under a tier-above Claude Opus 4.8 judge (Zheng et al., 2023); the lift is significant under GPT-5.5 ($p < 0.001$) but not Opus 4.8 ($p = 0.23$; §F)—directional rather than confirmed. The cost surfaces sharply once m2 is in the table: the raw-snapshot LLM issues *no* interventions, but adding the FM signal (m2→m6) makes it escalate—and 44.8% of those interventions misfire (FIR 0 → 0.448; $p < 10^{-4}$). The safety gate (m6→m7) reduces FIR only marginally (0.448 → 0.437), and its effects on F1 and judge score are within noise (§F). Both LLM variants reach grounding 1.000, so the failure mode is not citation hallucination but *over-reading the forecaster’s evidence as warrant for high-severity action*.¹

Judge robustness across model families. We re-judge m2/m6/m7 with a cross-family panel (Table 2). All three frontier judges rank m7 highest (mean score varying by ≤ 0.45). The one disagreement—Gemini 2.5 Pro scores m2 above m6 (2.69 vs 2.55), still below m7—sits within Gemini’s $\sim 3\times$ higher weekly noise (std 1.3–1.5 vs 0.3–0.6). Cross-family agreement on the top system closes the judge-family confound.

A tier-below, cheaper decision LLM dominates the headline operating point. We re-run the decision layer of m6 and m7 on the *same* FM-derived prompts with two non-Opus 4.7 decision LLMs—Claude Sonnet 4.6 (within-family, tier-below, $\sim 5\times$ cheaper) and Google Gemini 2.5 Pro (cross-family)—and re-score with the same Opus 4.8 judge (Table 3). On both architectures Sonnet 4.6 is a Pareto improvement over Opus 4.7: F1 rises (+15%/+24%; $p < 0.001$ on m7), judge qual-

ity is directionally higher (+0.11/+0.20; n.s.), and FIR collapses from 0.437–0.448 to 0.000—genuinely: Sonnet 4.6 still issues interventions at Opus-comparable volume (~ 190 vs 252), all on truly-stressed targets. Gemini 2.5 Pro is more conservative still (lower F1, FIR 0), so over-intervention is a property of the decision model, not of LLM decision layers per se. The operating point a practitioner would now deploy is m7 with Sonnet 4.6 (F1 0.0288, FIR 0.000, judge 2.65).

Layer-wise contribution. Holding the decision policy fixed at LLM and adding the DeXposure-FM signal lifts ticket F1 by 32% (0.0183 → 0.0241; $p < 0.001$); holding the predictor fixed at DeXposure-FM and swapping rules for the LLM lifts F1 by another 27% (0.0190 → 0.0241). The two layers therefore make additive, non-substitutable contributions (Figure 2 in §E). Adding the safety gate trades 3% F1 for judge changes within noise; its value is operational—every intervention clears two gates with auditable logs.

Matched-budget view. F1 conflates target quality with volume: m6/m7 emit 6.3–6.5 targets per week vs 4.9 for m2 and 2.1 for m1. Truncating every method to its top- k targets per week ($k \in \{1, 2, 3\}$; §F.3) separates the two effects: the LLM variants still roughly double the rules baseline’s precision@ k ($p \leq 0.0055$), but the FM-fed and raw-snapshot LLM variants become statistically indistinguishable. The FM’s contribution is thus not a sharper top of the ranking but support for a *larger* target set at undiminished precision; under Opus 4.7 the expanded tail is where over-intervention concentrates.

Ablation: load-bearing vs reserve components. Under clean 2025 data the confidence gate is load-bearing: removing it raises FIR from 0.000 to 0.429. The scenario engine is load-bearing for target extraction: skipping it collapses ticket precision from 0.600 to 0.000. The data-health gate and multi-horizon diversity are dormant on clean data but activate as reserves under degradation. Under severe feature/edge masking (80–98%) the data-health gate suppresses up to 60% of interventions that would otherwise produce false alarms (FIR range [0.27, 0.60] when disabled vs 0.000 when triggered); multi-horizon monitoring generates $\sim 4\times$ more alerts in the three crisis windows at unchanged precision. Per-regime ablation audits are in §E.6.

¹m3_evolvegc and m4_fm_only are predictor-only and do not emit tickets (forecast quality in §E.1); EvolveGCN trails persistence downstream, consistent with its ~ 220 weekly-snapshot training budget.

ID	Pred.	Decision	Prec.↑	Recall↑	F1↑	Judge↑	FIR↓	Ground.↑	Stabil.↑
m1	persist.	rules	0.720	0.004	0.0076	—	—	—	0.514
m2	—	LLM (no FM)	0.575	0.009	0.0183	2.24	—	1.000	0.532
m5	DeXposure-FM	rules	0.600	0.010	0.0190	—	—	—	0.257
m6	DeXposure-FM	LLM	0.570	0.012	0.0241	2.41	0.448	1.000	0.488
m7	DeXposure-FM	LLM + safety gate	0.580	0.012	0.0233	2.45	0.437	1.000	0.435

Table 1: Headline b5_decision comparison on the frozen 2025 test split ($n = 29$). Rows are read as a multi-objective trade-off rather than a single leaderboard: m1 is the conservative high-precision baseline; m2 is the pure-LLM baseline (no forecaster); m6/m7 are the FM-grounded LLM variants. Judge scores (1–5 scale) are from a Claude Opus 4.8 judge, strictly tier-above the Claude Opus 4.7 decision LLM; cross-family agreement is in Table 2. — in FIR: undefined— m1/m2/m5 emit no intervention-level ticket that could misfire.

Method	Opus 4.8	Gemini 2.5 Pro	GPT-5.5	Haiku 4.5 [†]
m2	2.24	2.69	1.90	2.03
m6	2.41	2.55	2.45	2.48
m7	2.45	2.90	2.45	2.69

Table 2: Cross-family judge panel on the same Opus 4.7 decisions ($n = 29$). Gemini and GPT use reasoning = {effort : minimal} so the thinking budget matches the Anthropic baseline. [†] Haiku 4.5 is a tier-below weak-judge reference.

Method	Decision LLM	F1↑	FIR↓	Judge↑
m6	Opus 4.7 (main)	0.0241	0.448	2.41
m6	Sonnet 4.6	0.0276	0.000	2.52
m6	Gemini 2.5 Pro	0.0129	0.000	2.21
m7	Opus 4.7 (main)	0.0233	0.437	2.45
m7	Sonnet 4.6	0.0288	0.000	2.65
m7	Gemini 2.5 Pro	0.0139	0.000	2.38

Table 3: Cross-decision-LLM panel on the same FM-derived prompts, judge fixed at Claude Opus 4.8 ($n = 29$). Sonnet 4.6 dominates Opus 4.7 on both architectures. The bold row (m7 $\times$ Sonnet 4.6) beats every row in Table 1 on every LLM-bearing axis at $\sim 5\times$ lower cost.

Early-warning timeline (historical event study).

For the three pre-test crisis windows the shared weighted-degree monitor (h1) achieves median lead times of 4–5 weeks with precision 1.000 at SVB/USDC across all alert budgets $K \in \{5, 10, 20\}$; per-budget details in §E.4. This is historical monitor evidence, not a test-split result.

7 Deployment Lessons

Three lessons transfer to other forecast→LLM pipelines.

L1. Grounding $\neq$ correctness. Every LLM variant we test achieves grounding 1.000: every cited number traces to a field of the evidence bundle. The FM-grounded variants still pay $\text{FIR} = 0.448/0.437$

(ungated/gated) under Opus 4.7. The failure is not invented evidence but *over-reading* of weak-but-present evidence as justification for high-severity action. Grounding certifies citations, not safety; only a downstream intervention-realism metric such as FIR does.

L2. Audit logs are a design artefact, not an output. Every ticket carries the forecast distribution, monitor scores, scenario CVaRs, rationale, gate states, and seeds that produced it, so a more conservative supervisor can retune τ_{data} or τ_{conf} and rerun the decision policy post-hoc, without re-invoking the FM or the LLM.

L3. Safety gates are reserves, not always-on filters. On clean 2025 data the data-health gate is dormant—invisible to a leaderboard without a degradation track—but under masking it suppresses up to 60% of FIR-eligible tickets: evaluate gates on dedicated stress regimes.

Limitations

Backtest, not field deployment. All results come from a historical backtest over 283 weekly snapshots of DeXposure (2020–2025). We have not run DeXposure-Claw as a live weekly cron at a supervisory authority. The ticket-level audit logs we release come from the backtest. Field deployment lessons—operator behaviour, end-to-end latency under real data feeds, regulator-in-the-loop calibration—are out of scope for this paper and remain the most pressing future direction.

Single domain. DeXposure-Bench evaluates a single on-chain risk surface (DeFi inter-protocol credit exposure). Generalisation to NFT lending, perpetual derivatives, or cross-chain bridge networks is untested and will require domain-specific re-calibration of the conformal split and the stressed-set percentile π .

Weekly resolution. The pipeline operates at weekly granularity, but several DeFi shocks unfold within hours—Terra/Luna erased \$40B in under 48 hours. A sub-hourly production deployment would require denser snapshot ingestion and re-evaluation of conformal calibration on the resulting denser time series. The 4–5 week lead times we report should be read as “how early the weekly monitor turns”, not as end-to-end alert latency.

Judge thinking-budget is held minimal. The cross-family judge panel (Claude Opus 4.8, Google Gemini 2.5 Pro, OpenAI GPT-5.5) is run with reasoning effort set to *minimal* so the comparison against the non-reasoning Anthropic baseline is a like-for-like comparison of intrinsic model capability rather than thinking budget. Default reasoning mode on Gemini 2.5 Pro and GPT-5.5 would change the reported scores; we treat the reasoning-on configuration as a distinct evaluation surface and leave it for a follow-up study. Practitioners deploying the judge at default reasoning settings should re-calibrate the score thresholds reported here. Human-judge calibration with practising supervisors remains the most pressing methodological extension.

Offline decision evaluation. `b5_decision` scores tickets against a regulator-aligned absolute-loss ground truth, which captures supervisory targeting priority but not the downstream effect of supervisory action—whether the recommended risk reduction would actually have averted contagion. A controlled live evaluation with a regulator in the loop, which DeXposure-Bench approximates but cannot replace, remains the primary direction we plan to extend the harness in.

Ethics Statement

Data provenance. DeXposure is built from public on-chain measurements aggregated via DefiLlama (DefiLlama, 2026). No private user data, wallet-level personally-identifying information, or off-chain identity data is used. Protocol-level aggregates are the unit of analysis throughout.

Decision risk. DeXposure-Claw emits supervisory tickets, not automated trades or automated interventions. The high-severity action types (Recommend–Reduce, Contingency) are gated on a data-health score and a confidence score, and the pipeline emits a safe-mode notice when either gate

fails. We characterise the system as decision support for a human supervisor, not as an autonomous risk agent. The measured false-intervention rate (FIR ≈ 0.44 for the DeXposure-FM+LLM variant) is reported in the headline results table so that downstream operators can weigh the trade-off against their own risk tolerance before deployment.

LLM use disclosure. The decision-layer LLM (`m2`, `m6`, `m7`) is Claude Opus 4.7 in the main configuration; Claude Sonnet 4.6 and Google Gemini 2.5 Pro are reported as decision-layer ablations. The LLM-as-judge for the explanation-quality axis is a three-model cross-family panel: Claude Opus 4.8 (within-family upgrade, strictly tier-above Opus 4.7), Google Gemini 2.5 Pro, and OpenAI GPT-5.5 (cross-family). All judges are run with reasoning effort set to *minimal* so the comparison is controlled for thinking budget. Claude Haiku 4.5 from a prior run is reported in the appendix as a weak-judge baseline. Every model is accessed through the OpenRouter unified API at temperature 0.

Release. We will release the harness, the eight reference implementations, and the ticket-level audit logs under a permissive open licence. Model checkpoints will be released subject to dataset-licence compatibility checks.

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A Forecaster Interface and Notation

A.1 Credit exposure

Let $X_G(q)$ be the set of tokens issued by protocol q and $X_p(t)$ the set of tokens held by protocol p at time t . Protocol p has a credit exposure to protocol q at t if $X_p(t) \cap X_G(q) \neq \emptyset$ (Wu et al., 2025). This token-mediated dependence generalises counterparty credit risk from traditional finance to DeFi.

A.2 Network representation

Over an interval $\tau = t_2 - t_1$, exposures form a weighted directed graph $G_\tau = (P_\tau, E_\tau)$. Node weights are $w_\tau(p) = \sum_{\sigma \in X_p(t_1) \cap X_p(t_2)} v_{t_2, \sigma}$ where $v_{t_2, \sigma}$ is the USD value of token σ at t_2 ; edge weights aggregate non-negative token flows from p to q mapped to issuer q , as defined in Wu et al. (2025). We adopt the same construction without modification.

A.3 Forecast primitive and predicted graph

At each decision epoch t the agent queries $\mathcal{P}_{t,h} \leftarrow \text{DeXposure-FM.Forecast}(G_t, X_t, h)$. From $\mathcal{P}_{t,h}$ we build a representative predicted graph via a thresholded expected-weight operator,

$$\hat{w}_{pq,t+h} := \Pr(e_{pq,t+h} = 1) \cdot \mathbb{E}[w_{pq,t+h} \mid e_{pq,t+h} = 1], \quad (19)$$

$$\hat{E}_{t+h} := \{(p, q) : \Pr(e_{pq,t+h} = 1) > \pi_{\min}\}, \quad (20)$$

with $\pi_{\min} = 0.2$ (tuned on validation). For uncertainty quantification we draw $M = 50$ Monte Carlo samples $\{\tilde{G}_{t+h}^{(m)}\}$ by Bernoulli edge sampling and Gaussian weight perturbation of $\mathcal{P}_{t,h}$.

A.4 Hybrid prediction strategy

In practice DeXposure-FM keeps all edges from G_t , reweights them as $w'_{pq} = w_{pq} + \Pr(e_{pq,t+h} = 1) \cdot \hat{r}_{pq}$ where $\hat{r}_{pq}$ is the FM residual, and additionally adds new edges whose existence probability exceeds $\pi_{\min}$. This preserves network structure while letting the FM adjust weights and predict new connections.

A.5 Architecture and training

DeXposure-FM is built on the GraphPFN graph-tabular foundation backbone (Hollmann et al., 2025; Ereemeev et al., 2025) with a LiMiX-16M tabular transformer encoder: 12 layers, 4 attention heads, hidden dimension 64. The same backbone serves three heads—edge existence (binary), edge weight (regression), and node-TVL change (regression)—trained jointly with a BCE+MSE composite loss. We fine-tune one checkpoint per horizon $h \in \{1, 4, 8, 12\}$ for 20 epochs with AdamW (learning rate 10^{-4} for heads, 10^{-5} for the backbone). Fine-tuning draws on DeXposure snapshots from March 2020 to January 2025: 104 training weeks, 12 validation weeks (used for early stopping), and 8 subsequent held-out weeks for internal evaluation. All data that influences weights or checkpoint selection precedes the frozen 2025 leaderboard split (§C). Full-graph in-context inference fits a single RTX 4090; one weekly forecast across all four horizons completes in well under a minute.

Held-out link-prediction quality is AUPRC = 0.978 / 0.973 / 0.967 / 0.967 and AUROC = 0.996 / 0.995 / 0.993 / 0.993 at $h = 1 / 4 / 8 / 12$ respectively; a frozen GraphPFN baseline (no fine-tuning) achieves AUPRC 0.936–0.940 at the same horizons, isolating the contribution of domain fine-tuning. These intrinsic forecaster numbers do not appear in the main text because the paper’s contribution is the operating point of the full pipeline, not the FM’s headline accuracy.

B Pipeline Mathematics

B.1 Data-health gate

The data-health score combines four deterministic checks of the incoming snapshot:

$$DH_t = \text{mean}(\text{fresh}_t, \text{miss}_t, \text{topo}_t, \text{disc}_t) \in [0, 1]. \quad (21)$$

Safe mode is triggered when $DH_t < \tau_{\text{data}} = 0.7$: monitor alerts are still emitted but intervention-type tickets are suppressed.

B.2 Monitor metrics

ID	Metric	Definition
N1	Systemic Importance	$\max_{p \in V} \text{PageRank}(\hat{G}_{t+h})_p$
N2	HHI Concentration	$\sum_p (d_p / \sum_q d_q)^2$, d_p weighted out-degree
N3	Network Density	$ E_{t+h} / V (V - 1)$
N4	PageRank Gini	$\text{Gini}(\{\text{PR}_p\}_{p \in V})$
N5	Degree Gini	$\text{Gini}(\{d_p\}_{p \in V})$

PageRank uses power iteration with damping 0.85 and 10 iterations. An alert fires when $|N_{t,h}^{(j)} - \mu_t^{(j)}| / \max(\sigma_t^{(j)}, \epsilon) > z$ with $z = 2$, $L = 42$ week rolling baseline, $\epsilon = 10^{-8}$. The b2_warning historical event study uses a shorter 26-week (≈ 6 -month) baseline window.

B.3 Uncertainty-aware alert confidence

With $\text{CV}_{t,h}^{(j)}$ the coefficient of variation of metric j across the M MC samples (clamped to $[0, 1]$),

$$C_t(a) := DH_t \cdot \frac{1}{1 + \text{CV}_{t,h}^{(j)}} \cdot \frac{1}{1 + \ln(1 + h)} \in [0, 1]. \quad (22)$$

$C_t(a)$ decreases monotonically with lower data health, wider predictive spread, and longer horizons.

B.4 Scenario engine

ID	Scenario	Target	Shock
S1	Single protocol fail	Highest-weight node	100%
S2	Bridge cluster fail	All bridge nodes	100%
S3	Stablecoin de-peg	All stablecoin nodes	50%
S4	Sector lending shock	All lending nodes	30%
S5	Correlated stress	Top-10 by weight	20%

Per-scenario loss $L_{t,h}(s) = (\sum_e w_e(\hat{G}_{t+h}) - \sum_e w_e(\hat{G}_{t+h}^s)) / \sum_e w_e(\hat{G}_{t+h})$; CVaR aggregation averages the worst $\lceil \lambda M \rceil$ MC-sample losses with $\lambda = 0.5$.

B.5 Decision feasibility and ticket scoring

An intervention-level action u is feasible iff

$$(\text{SAFE_MODE}_t = 0) \wedge (\bar{C}_t \geq \tau_{\text{conf}} = 0.6), \quad (23)$$

with $\bar{C}_t$ the mean alert confidence. Feasible tickets are scored as $\text{Score}_t(u) = w_u \cdot \bar{C}_t \cdot \text{Imp}_t$, where $w_u \in \{0.25, 0.50, 0.75, 1.00\}$ is the playbook severity weight and Imp_t is the mean CVaR across scenarios (default 1.0 when no scenarios are available).

B.6 End-to-end procedure

C Benchmark Details

C.1 Six-axis schema

ID	Capability tested	Primary metrics
b1_forecast	Temporal graph prediction	PageRank/HHI/Gini MAE, trend consistency, Spearman; per $h \in \{1, 4, 8, 12\}$
b2_warning	Streaming anomaly detection	Precision, recall, F1, lead time, alert stability; per event $\times$ budget
b3_calibration	Predictive uncertainty	PI coverage at 0.90 target, PI width, ECE, CRPS
b4_stress	What-if scenario fidelity	Loss MAE, distressed-count MAE, propagation depth MAE, target overlap@ k
b5_decision	Supervisory ticket quality	Precision, recall@ k , F1, target stability, judge score, FIR
b6_robustness	Data quality sensitivity	Per-benchmark metrics under 5 degradation regimes; relative degradation

The six axes are chosen to be (i) *non-overlapping*—a system can excel at b1 while failing b5; (ii) *decomposable along the four-layer framework*, so that ablations on the predictor layer (b1, b3, b6), monitor layer (b2), scenario layer (b4), and decision layer (b5) can be reported independently; and (iii) *individually publishable*, so that downstream work may extend a single axis without re-running the whole suite.

Algorithm 1 DeXposure-Claw: one decision epoch.

Require: (G_t, X_t) ; horizons $\mathcal{H}$; scenarios $\mathcal{S}_0$; thresholds $(\tau_{\text{data}}, \tau_{\text{conf}}, z, \pi_{\text{min}})$; MC count M

Ensure: Alerts $\mathcal{A}_t$; scenario summary $\mathcal{R}_t$; tickets $\mathcal{D}_t$

```
1:  $DH_t \leftarrow \text{DataHealth}(G_t, X_t)$ ;  $\text{SAFE\_MODE}_t \leftarrow \mathbb{I}\{DH_t < \tau_{\text{data}}\}$ 
2: for all  $h \in \mathcal{H}$  do
3:    $\mathcal{P}_{t,h} \leftarrow \text{DeXposure-FM.Forecast}(G_t, X_t, h)$ 
4:    $\hat{G}_{t+h} \leftarrow \text{BuildPredGraph}(\mathcal{P}_{t,h}; \pi_{\text{min}})$ 
5:    $\{\tilde{G}_{t+h}^{(m)}\} \leftarrow \text{MCSample}(\mathcal{P}_{t,h})$ 
6:   Compute  $\mathbf{N}_{t,h} = \Phi(\hat{G}_{t+h}, X_t)$ , attach rolling stats, derive alerts  $\mathcal{A}_{t,h}$  and confidences  $C_t(\cdot)$ 
7:   for all  $s \in \mathcal{S}_0$  do
8:     Compute  $L_{t,h}(s), \tilde{L}_{t,h}^{(m)}(s), \text{CVaR}_{0.5}(s, h)$ 
9:   end for
10: end for
11:  $\mathcal{D}_t \leftarrow \text{PlaybookDecision}(\mathcal{A}_t, \mathcal{R}_t, DH_t, \text{SAFE\_MODE}_t, \tau_{\text{conf}})$ 
12: return  $(\mathcal{A}_t, \mathcal{R}_t, \mathcal{D}_t)$ 
```

C.2 Dataset split

Split	Period	Weeks	Usage
Train	2020-03–2024-06	~222	Forecaster training
Validation	2024-07–2024-12	~26	Conformal calibration, threshold tuning
Test	2025-01–2025-08	~33	Frozen leaderboard

C.3 Ground-truth equation

With $w_t(v) = \sum_{e \ni v} w_t(e)$ the total weight incident on v , the regulator-aligned stressed set at horizon h is

$$\Delta_t^h(v) = w_t(v) - w_{t+h}(v), \quad \mathcal{S}_t^h = \text{top}_\pi \{v : w_t(v) > 0, \Delta_t^h(v) > 0\}, \quad (24)$$

with $\pi = 0.05$ and ground-truth horizon $h = 4$ weeks. The final four test weeks lack the 4-week-ahead snapshot w_{t+4} and are not scored, leaving $n = 29$ evaluated weeks. This is the single ground-truth definition used by b2, b5, and the LLM-decision evaluation pipeline.

C.4 Reporting protocol

A submission produces one JSON file per {benchmark, method} pair following the harness schema, plus a single results.json aggregating the suite. The full leaderboard requires b1, b3, b5, b6; b2 and b4 are required only for systems that implement a monitor or scenario layer respectively. The harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files.

D Reference Methods

ID	Predictor	Policy	Role
h1_weighted_degree	—	heuristic alert	Streaming anomaly baseline (b2)
m1_persistence_rules	$G_{t+h} = G_t$	rule engine	Decision baseline
m2_snapshot_llm	current snap	LLM	LLM-only ablation (no forecast)
m3_evolvegcnn	EvolveGCN	—	Learned-GNN baseline
m4_fm_only	DeXposure-FM	—	FM-only ablation
m5_fm_rules	DeXposure-FM	rule engine	FM + rules
m6_fm_llm	DeXposure-FM	LLM	Full FM + LLM stack
m7_fm_llm_gated	DeXposure-FM	LLM + safety gate	Safety-gated FM + LLM

The eight methods expose all four cuts of the four-layer ablation: adding the predictor layer (m1→m5), swapping the decision layer (m5→m6), adding the safety gate (m6→m7), and removing the predictor from the LLM stack (m2→m6). All methods conform to a shared predict_graph/decide_actions interface; adding a ninth method is a single-file change.

LLM model panel and decoding. All LLM-bearing methods (m2, m6, m7) are evaluated under three decision-layer configurations: Claude Opus 4.7 (main), Claude Sonnet 4.6 (within-family, tier-below ablation), and Google Gemini 2.5 Pro (cross-family ablation). Each runs at temperature 0 and

max_tokens = 4096; self-consistency is probed with three repeated calls per week and Jaccard overlap on emitted ticket targets and severities.

The LLM-as-judge for the explanation-quality axis is a three-model panel chosen so the judge tier is at or above the decision tier on the public 2026 Artificial Analysis index: Claude Opus 4.8 (within-family upgrade, strictly tier-above Opus 4.7), Google Gemini 2.5 Pro, and OpenAI GPT-5.5 (cross-family upgrades). The Gemini and GPT judges are reasoning-by-default models; we attach OpenRouter’s reasoning = {effort : minimal} control to both so the judge thinking budget matches the non-reasoning Anthropic baseline. Claude Haiku 4.5 from a pilot run is reported as a weak-judge reference. Every model is served through the OpenRouter unified API; observed cost is on the order of a few US dollars per full sweep of the 29-week test split.

E Full Per-Slice Results

This appendix expands the headline numbers of §6 into the per-slice audits underlying them.

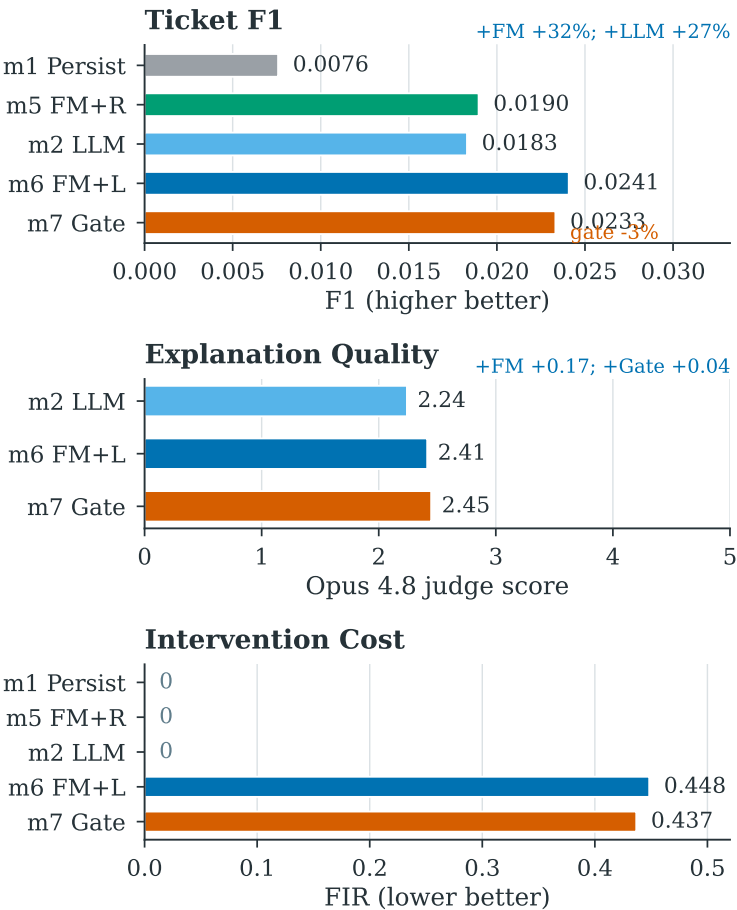


Figure 2: Layer-wise contribution and cost (visualising Tables 1 and 3). Adding the FM signal to a snapshot LLM improves ticket F1 and explanation quality; swapping the FM-fed rule engine for the LLM improves recall-oriented F1 further; the same LLM variants incur a substantial false-intervention rate, partially mitigated by the safety gate.

Table 4: b1_forecast all-horizon forecasting on the 2025 test split. PageRank MAE shown in 10^{-5} units. Best per horizon and metric in **bold**; ties share the bold. Trend consistency for m1_persistence is structurally 0 because $\hat{G}_{t+h} = G_t$ produces no trend signal.

h	Method	PR MAE ($\downarrow$)	HHI MAE ($\downarrow$)	Gini MAE ($\downarrow$)	Rank Corr ($\uparrow$)	Trend Cons ($\uparrow$)
1	m1_persistence	3.6	0.0376	0.0027	0.556	0.000
	m3_evolvegc	3.7	0.0481	0.1149	0.535	0.444
	m4_fm_only	4.7	0.0376	0.0029	0.549	0.531
	m5_fm_rules	4.7	0.0376	0.0029	0.549	0.525
4	m1_persistence	3.4	0.0322	0.0018	0.570	0.000
	m3_evolvegc	3.5	0.0489	0.1326	0.551	0.324
	m4_fm_only	4.5	0.0321	0.0022	0.559	0.621
	m5_fm_rules	4.5	0.0321	0.0022	0.558	0.628
8	m1_persistence	3.9	0.0399	0.0028	0.517	0.000
	m3_evolvegc	4.0	0.0492	0.1140	0.499	0.296
	m4_fm_only	4.5	0.0398	0.0030	0.508	0.624
	m5_fm_rules	4.5	0.0398	0.0030	0.508	0.632
12	m1_persistence	3.7	0.0393	0.0027	0.525	0.000
	m3_evolvegc	3.8	0.0534	0.1245	0.509	0.314
	m4_fm_only	4.1	0.0392	0.0029	0.516	0.600
	m5_fm_rules	4.1	0.0392	0.0029	0.516	0.600

Table 5: b4_stress per-scenario detail. Loss MAE in normalised units; Distress count MAE in protocols; target overlap@10 measured against the actual future graph. Best per scenario and metric in **bold**; ties share the bold, and columns where all four methods tie are left unbolded. The main pattern is conservative: FM variants roughly match the strong persistence baseline on loss and target-overlap metrics, while EvolveGCN is generally weaker on downstream what-if fidelity.

Scenario	Method	Loss MAE ($\downarrow$)	Distress MAE ($\downarrow$)	Prop. Depth MAE ($\downarrow$)	Overlap@10 ($\uparrow$)
S1 (single protocol)	m1_persistence	0.0668	200.3	0.897	0.314
	m3_evolvegc	0.1072	218.1	0.966	0.034
	m4_fm_only	0.0667	201.4	0.897	0.314
	m5_fm_rules	0.0667	201.6	0.897	0.314
S2 (bridge cluster)	m1_persistence	0.0150	27.3	0.000	0.521
	m3_evolvegc	0.0116	196.4	0.000	0.189
	m4_fm_only	0.0151	50.3	0.000	0.525
	m5_fm_rules	0.0151	52.1	0.000	0.525
S3 (stablecoin de-peg)	m1_persistence	0.0079	0.0	0.000	0.445
	m3_evolvegc	0.0156	0.0	0.000	0.269
	m4_fm_only	0.0080	0.0	0.000	0.445
	m5_fm_rules	0.0080	0.0	0.000	0.445
S4 (sector lending)	m1_persistence	0.0100	0.0	0.034	0.536
	m3_evolvegc	0.0200	0.0	0.034	0.293
	m4_fm_only	0.0099	0.0	0.034	0.536
	m5_fm_rules	0.0099	0.0	0.034	0.536
S5 (correlated top-10)	m1_persistence	0.0100	0.0	0.000	0.602
	m3_evolvegc	0.0538	0.0	1.000	0.330
	m4_fm_only	0.0099	0.0	0.000	0.602
	m5_fm_rules	0.0099	0.0	0.000	0.602

Table 6: b6_robustness per-regime detail. All metrics at $h=4$ on the 2025 test split under five degradation regimes. PageRank, HHI, and Gini MAE in 10^{-3} units. Relative degradation is signed against the clean b1_forecast baseline; negative values mean the predictor is no worse than on clean data. Best per metric (lowest MAE, highest rank correlation, smallest $|\Delta_{\text{rel}}|$) in **bold**; ties share the bold. Persistence ignores node features, so its noisy_features_01 and missing_features_20 rows degenerate to the clean numbers (and $\Delta_{\text{rel}} = 0$); EvolveGCN’s predictions also change only negligibly under these two regimes ($|\Delta_{\text{rel}}| < 10^{-8}$).

Regime	Method	PR MAE ($\downarrow$)	HHI MAE ($\downarrow$)	Gini MAE ($\downarrow$)	Rank Corr ($\uparrow$)	Δ_{rel} (~ 0)
low_data_10pct	m1_persistence	15.94	0.0591	0.0023	0.552	+0.766
	m3_evolvegc	15.23	0.0999	0.1301	0.535	+ 0.276
	m4_fm_only	12.83	0.0586	0.0027	0.528	+0.612
	m5_fm_rules	12.71	0.0587	0.0027	0.528	+0.607
low_data_25pct	m1_persistence	9.46	0.0282	0.0020	0.577	−0.093
	m3_evolvegc	10.22	0.0486	0.1281	0.558	− 0.027
	m4_fm_only	9.86	0.0279	0.0024	0.553	−0.126
	m5_fm_rules	9.95	0.0279	0.0024	0.553	−0.126
partial_graph_30	m1_persistence	9.63	0.0350	0.0019	0.554	+0.068
	m3_evolvegc	10.36	0.0488	0.1236	0.537	− 0.048
	m4_fm_only	12.52	0.0350	0.0022	0.500	+0.082
	m5_fm_rules	12.45	0.0350	0.0022	0.499	+0.077
noisy_features_01	m1_persistence	9.77	0.0322	0.0018	0.584	0.000
	m3_evolvegc	10.68	0.0489	0.1326	0.563	0.000
	m4_fm_only	11.38	0.0323	0.0021	0.554	−0.005
	m5_fm_rules	11.35	0.0323	0.0021	0.554	−0.007
missing_features_20	m1_persistence	9.77	0.0322	0.0018	0.584	0.000
	m3_evolvegc	10.68	0.0489	0.1326	0.563	0.000
	m4_fm_only	12.11	0.0323	0.0023	0.559	+0.017
	m5_fm_rules	12.08	0.0323	0.0023	0.559	+0.014

Table 7: b2_warning per-event detail for the shared weighted-degree monitor (h1_weighted_degree) across alert budgets $K \in \{5, 10, 20\}$. Recall and F1-warning shown in 10^{-3} units (the full DeFi corpus contains $\sim 4,300$ protocols, so per-week recall denominators are large). Lead time is event-level and not budget-sensitive: 5 weeks for Terra/Luna and FTX, 4 weeks for SVB-USDC. b2_warning sits upstream of the predictor choice, so a single monitor is shared across all methods (see §6).

Event	K	Prec. ($\uparrow$)	Rec. $\times 10^3$ ($\uparrow$)	F1 _w $\times 10^3$ ($\uparrow$)	Stab. ($\uparrow$)	$ \mathcal{A} $
Terra/Luna	5	0.60	1.81	3.61	0.44	11
	10	0.70	3.26	6.49	0.62	19
	20	0.65	7.97	15.74	0.74	36
FTX	5	0.60	1.33	2.65	0.64	10
	10	0.70	2.39	4.77	0.72	18
	20	0.55	4.79	9.49	0.72	39
SVB-USDC	5	1.00	1.81	3.62	0.55	10
	10	1.00	2.82	5.62	0.75	15
	20	1.00	5.64	11.22	0.81	30

Table 8: Per-regime data-health gate ablation isolating A1 from A2. Each row holds $\tau_{\text{conf}}=0$ and sweeps τ_{data} . “strict” = $\tau_{\text{data}}=0.85$, “on” = 0.70, “off” = 0.00. The four regimes apply targeted edge or feature degradation on the clean 2025 test split ($n_{\text{weeks}}=29$): extreme masks both topology and features at 90%, topo removes 90% of edges only, feat corrupts 90% of node features only, severe masks both at 80%. Numbers expand Panel B of Table 9: under degradation the gate-off setting produces FIR up to 0.541, while *on/strict* can suppress intervention tickets via safe-mode and drive FIR to 0.

Regime	Gate	τ_{data}	$\overline{d\bar{h}}$	Safe-mode (%)	Ticket Prec. ($\uparrow$)	FIR ($\downarrow$)	Interventions
extreme	strict	0.85	0.587	100	0.510	0.000	0
	on	0.70	0.587	100	0.545	0.000	0
	off	0.00	0.587	0	0.490	0.510	57
extreme_topo	strict	0.85	0.637	100	0.517	0.000	0
	on	0.70	0.637	100	0.462	0.000	0
	off	0.00	0.637	0	0.559	0.441	58
extreme_feat	strict	0.85	0.755	100	0.503	0.000	0
	on	0.70	0.755	0	0.510	0.267	5
	off	0.00	0.755	0	0.503	0.267	5
severe	strict	0.85	0.751	100	0.490	0.000	0
	on	0.70	0.751	0	0.538	0.400	25
	off	0.00	0.751	0	0.538	0.541	24

E.1 Full all-horizon b1_forecast results

E.2 Per-scenario b4_stress results

E.3 Per-regime b6_robustness results

E.4 Per-budget b2_warning results

E.5 Per-regime A1 (data-health gate) isolation

E.6 Stress-condition ablation Panels B and C

Panel B: under severe feature/edge masking, FIR ranges over $[0.27, 0.60]$ when the data-health gate is disabled ($\tau_{\text{data}} = 0$; the confidence gate is held off at $\tau_{\text{conf}} = 0$ throughout Panel B to isolate A1), and collapses to 0.000 when the gate is triggered, by suppressing intervention tickets. Panel C: multi-horizon monitoring ($h \in \{1, 4, 8, 12\}$) generates 73–98 alerts in the three crisis windows vs 19–23 for single-horizon ($h = 4$), at unchanged precision range 0.533–0.624 and zero false interventions in both cases.

F Statistical Significance and Matched-Budget Analysis

All headline metrics are means over the $n = 29$ scored test weeks. This appendix reports week-level uncertainty for the central comparisons: 95% percentile bootstrap confidence intervals (10,000 resamples of weeks with replacement; F1 is recomputed from the resampled mean precision and recall) and two-sided paired sign-flip permutation tests (20,000 permutations, methods aligned by week). Analysis code is released with the harness (scripts/bootstrap_stats.py, scripts/matched_budget.py).

F.1 Confidence intervals

F.2 Paired permutation tests

Three readings follow. (i) The load-bearing quantitative claims—the FM signal’s F1 lift, the full stack’s F1 lift over rules, the Sonnet 4.6 F1 improvement, and the over-intervention cost—are significant at $p < 0.001$. (ii) The explanation-quality lift of the FM-grounded variants is judge-dependent: significant under GPT-5.5, positive but not significant under Opus 4.8 ($p = 0.23$), and absent under the (noisiest) Gemini judge. We therefore read the judge axis as directional evidence rather than a confirmed effect. (iii) The safety gate’s effects on F1 and judge score are indistinguishable from noise on clean data, consistent with its characterisation as a reserve mechanism (§7).

Table 9: Component-level ablation of DeXposure-Claw on b5_decision. Panel A shows clean 2025 test-split ablations against the m5_fm_rules baseline; Panels B–C report targeted stress runs for mechanisms that are dormant on clean data. A2 (confidence gating) and A3 (scenario engine) are load-bearing in the clean setting, while A1 (data-health gating) and A6 (multi-horizon monitoring) are safety-reserve mechanisms whose effects appear under data degradation and crisis-window dynamics, respectively.

Panel	Ablation	Configuration	Ticket Prec.	FIR	Notes
A: clean test	—	full m5_fm_rules	0.600	0.000	baseline
	A1	disable data-health gate ($\tau_{\text{data}}=0$)	0.600	0.000	no effect (clean data)
	A2	disable confidence gate ($\tau_{\text{conf}}=0$)	0.600	0.429	FIR shoots up
	A3	skip scenario engine	0.000	0.000	target extraction collapses
	A6	single-horizon forecasting ($h=4$)	0.600	0.000	no effect (clean data)
B: degraded data	A1 off	80–98% feature/edge mask, $\tau_{\text{conf}}=0$	—	0.27–0.60	false interventions emerge
	A1 on	same degradation, safe-mode triggered	—	0.000	intervention suppression
C: crisis period	A6: single	horizons = {4}, crisis windows	0.533–0.624	0.000	alerts: 19–23
	A6: multi	horizons = {1, 4, 8, 12}, same windows	0.533–0.624	0.000	alerts: 73–98 ($\approx 4\times$)

Table 10: 95% bootstrap CIs for the decision-quality headline numbers (decision LLM in parentheses; judge is Claude Opus 4.8 throughout). [‡]m1’s per-week distribution comes from a local re-run of the released harness: its aggregate F1 (0.0081) brackets the published GPU-run value (0.0076), and the recall@ k and stressed-pool size reproduce exactly, but with <1 ticket per week a few borderline tickets flip across environments (re-run precision 0.633 vs published 0.720), so the m1 interval should be read as order-of-magnitude.

Configuration	F1 [95% CI]	FIR [95% CI]	Judge [95% CI]
m1 rules [‡]	0.0081 [0.0041, 0.0123]	0.000	—
m2 (Opus 4.7)	0.0184 [0.0142, 0.0223]	0.000	2.24 [2.10, 2.41]
m6 (Opus 4.7)	0.0241 [0.0201, 0.0281]	0.448 [0.293, 0.601]	2.41 [2.21, 2.62]
m7 (Opus 4.7)	0.0234 [0.0194, 0.0271]	0.437 [0.282, 0.592]	2.45 [2.24, 2.66]
m6 (Sonnet 4.6)	0.0276 [0.0225, 0.0324]	0.000	2.52 [2.28, 2.79]
m7 (Sonnet 4.6)	0.0288 [0.0236, 0.0335]	0.000	2.66 [2.41, 2.90]
m6 (Gemini 2.5 Pro)	0.0129 [0.0103, 0.0155]	0.000	2.21
m7 (Gemini 2.5 Pro)	0.0139 [0.0111, 0.0167]	0.000	2.38

Table 11: Two-sided paired permutation p -values for the key comparisons. Significant effects ($p < 0.05$) in **bold**.

Comparison	Metric	p
m6 vs m2 (FM signal)	F1	0.0002
m7 vs m1 (full stack)	F1	<0.0001
m7 vs m6 (gate)	F1	0.41
m7 $\times$ Sonnet vs m7 $\times$ Opus	F1	0.0004
m6 vs m2	FIR	<0.0001
m6 vs m2 (Opus 4.8 judge)	Judge	0.23
m6 vs m2 (GPT-5.5 judge)	Judge	0.0002
m6 vs m2 (Gemini judge)	Judge	0.77
m7 vs m6 (Opus 4.8 judge)	Judge	1.00
m7 $\times$ Sonnet vs m7 $\times$ Opus	Judge	0.21

F.3 Matched-budget recall@ k

Methods emit different numbers of ticket targets per week (m1 2.1, m2 4.9, m7 6.3, m6 6.5, m7 $\times$ Sonnet 7.6), and unbudgeted recall mechanically favours methods that flag more protocols. To separate target *quality* from target *volume*, we truncate every method to its top- k targets per week (LLM variants ranked by their own emitted risk score; m1 by ticket confidence) and score against the same stressed pool.

The matched-budget view decomposes the headline F1 result into two distinct effects. Against the *rules baseline*, LLM targeting is genuinely better per target: at every k , precision@ k roughly doubles

Table 12: Matched-budget recall@ k ($\times 10^3$) and precision@ k on the 2025 test split. At matched budgets every LLM variant beats the rules baseline ($p \leq 0.0055$ vs m1 at all k), while the FM-grounded and raw-snapshot LLM variants are statistically indistinguishable from each other (all pairwise $p \geq 0.10$) and the top- k hit patterns of the three FM-fed decision LLMs nearly coincide. m1 row from the local re-run (Table 10 note).

Method	$k=1$		$k=2$		$k=3$	
	R@1	P@1	R@2	P@2	R@3	P@3
m1 rules	0.71	0.24	1.71	0.28	2.45	0.26
m2 (Opus 4.7)	1.53	0.52	3.47	0.55	5.70	0.59
m6 (Opus 4.7)	1.64	0.55	3.29	0.53	5.30	0.56
m7 (Opus 4.7)	1.64	0.55	3.47	0.55	5.30	0.56
m7 (Sonnet 4.6)	1.64	0.55	3.47	0.55	5.30	0.56

(0.52–0.59 vs 0.24–0.28) and recall@ k is significantly higher ($p \leq 0.0055$). Against the *raw-snapshot LLM*, however, the FM’s contribution is not a higher per-target hit rate at the top of the ranking—at matched k the FM-fed and raw-snapshot variants are statistically indistinguishable, and the FM-fed variants’ top- k targets are nearly identical across decision LLMs because the evidence bundle pins the highest-exposure protocols. The FM’s F1 gain comes from supporting a *larger* target set at undiminished precision (Table 1), which is also where the over-intervention risk concentrates: under Opus 4.7 the expanded tail is escalated to intervention-level severity (FIR = 0.448), while Sonnet 4.6 expands the same set without escalating it (FIR = 0).

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